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# TBA4245 Geodesy

## Example exercise no. 7: GPS and troposphere

Using the formulas (5.107) and (5.120) GNSS book by Hofmann-Wellenhof, the aim is to obtain some numerical values of the tropospheric path delay.

**(a.1) Hopfield model, formula (5.107)**
Use the following average values for the observation site (the GPS receiver:

| | |
|---|---|
| Air temperature: | $t = 15°$ |
| Air pressure: | $p = 1013{,}25$ hPa |
| Water vapour, partial pressure: | $e = 20$ hPa |

Calculate the dry and wet delays through the troposphere for some different zenith angles between a GPS receiver and a GPS satellite:

$z = 0°$ $z = 85°$ $z = 90°$

Calculate the total tropospheric path delay.

**(a.2) Saastamainen model, formula (5.120)**

Same calculations as in (a.1)

**(b)**
Assume an uncertainty of the calculate value of 2% (2-5%) on the dry part and 10% (10-20%) on the wet part.

Calculate the uncertainty (in meters) of the dry and wet part of the tropospheric path delays.

Evaluate in what effect differential measurements eliminates or reduces the tropospheric path delays.

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Figure from a GPS textbook (Kaplan?), tropospheric delay vs elevation angle, E:

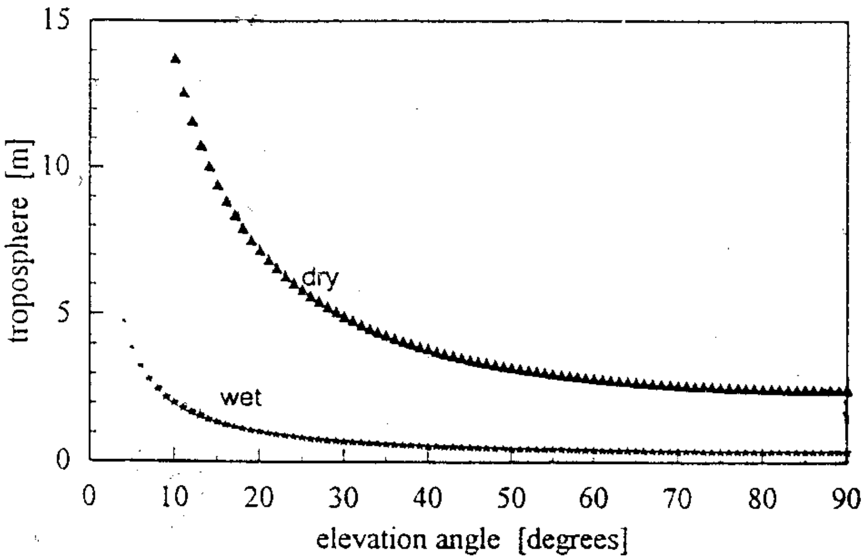


**Figure 9.7** Dry and wet tropospheric model delays.

## **Solution:**

(a.1)

Using the formulas (5.107) in the calculations

$$\text{Dry:}\quad \Delta_d^{Trop}(E)=\frac{10^{-6}}{5}\cdot\frac{77,64\cdot\frac{p}{T}}{\sin\left(\sqrt{E^2+6,25}\right)}\cdot\left[40136+148,72\cdot(T-273,16)\right]$$

$$\text{Wet:}\quad \Delta_w^{Trop}(E)=\frac{10^{-6}}{5}\cdot\frac{-12,96\cdot T+3,718\cdot 10^5}{\sin\left(\sqrt{E^2+2,25}\right)}\cdot\frac{e}{T^2}\cdot 11000$$

$E$ is the elevation angle (height angle): $E = 90° – Z$ , where Z is the zenith angle.
T = 273,16° is the absolute zero point for the temperature (the zero point of the Kelvin temperature scale).

$T$ is given in Kelvin, thus $T$ - 273,16° is in Celsius: $t$ = 15°C => $T$ = 288,16° K
Pressure in millibar (1013,25 mbar = 760 mm Hg = 1013,25 hPascal = standard atmosphere

Dry:

$$\Delta_d^{Trop}(E=0)=\frac{10^{-6}}{5}\cdot\frac{77,64\cdot\frac{1013,25}{288,16}}{\sin\left(\sqrt{0^2+6,25}\right)}\cdot\left[40136+148,72\cdot(288,16-273,16)\right]=53,0\text{ m}$$

$\Delta_d^{Trop}(E=5)=23,7\text{ m}$ $\qquad$ $\Delta_d^{Trop}(E=90)=2,31\text{ m}$

Wet:

$$\Delta_w^{Trop}(E=90)=\frac{10^{-6}}{5}\cdot\frac{-12,96\cdot 288,16+3,718\cdot 10^5}{\sin\left(\sqrt{90^2+2,25}\right)}\cdot\frac{20}{288,16^2}\cdot 11000=0,20\text{ m}$$

$\Delta_w^{Trop}(E=5)=2,14\text{ m}$ $\qquad$ $\Delta_w^{Trop}(E=0)=?$ m, not calculated

Check that the results are ca. the same as on the figure in the assignment text.

(a.2) Not calculated. Different models will give ca. the same results when the height angle $E$ > 15°, and normally they will be different when the height angle $E$ is < 15°.

(b)

| | $Z$ = 0° $E$ = 90° | | $Z$ = 85° $E$ = 5° | | $Z$ = 90° $E$ = 0° | |
|---|---|---|---|---|---|---|
| | Error | Uncertainty | Error | Uncertainty | Error | Uncertainty |
| Dry | 2,31 | 0,046 m | 23,7 | 0,474 m | 53,0 m | 1,06 m |
| Wet | 0,20 | 0,020 m | 2,14 | 0,214 m | | |

Even if the wet error is ca. 10 larger than the error on the dry part, we see from the uncertainty of the calculation of the error on the wet part that the uncertainties are of ca. the same size for the wet and dry part. This is because of the dry part is more stable and thus easier to model well.

Differential / relative observations (two phase receivers) together with modelling will eliminate most of the type of error, etc…